INCH-POUND
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SUPERSEDING
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15 December 1965
(See 6.8)

MILITARY SPECIFICATION

COOLERS, FLUID, NAVAL SHIPBOARD (HULL COOLERS)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hull mounted engine jacket water coolers for Naval shipboard application.

1.2 Classification. Coolers shall be of the following types and sizes (see 6.2). Unless otherwise specified (see 6.2), coolers shall be type A.

Type A
Type B
Size 100X
Size 150
Size 300X
Size 301
Size 400
Size 500
Size 1000

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-A-200	- Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded; General Specification for.
QQ-A-200/5	- Aluminum Alloy 5086, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-225	- Aluminum and Aluminum Alloy Bar, Rod, Wire, or Special Shapes; Rolled, Drawn, or Cold Finished; General Specification for.
QQ-A-225/7	- Aluminum Alloy 5052, Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished.
QQ-B-750	- Bronze, Phosphor; Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

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MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-G-1149	- Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness.
MIL-S-1222	- Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws and Nuts.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-T-15005	- Tubes, 70-30 and 90-10 Copper Nickel Alloy, Condenser and Heat Exchanger.
MIL-C-15726	- Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire.
MIL-T-16420	- Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).
MIL-A-18001	- Anodes, Corrosion Preventive, Zinc; Slab Disc and Rod Shaped.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-H28/2	- Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads - UN and UNR Thread Forms.

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MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.
MIL-STD-1399, Section 301 - Interface Standard for Shipboard Systems, Ship Motion and Attitude. (Metric)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials. The materials used in the construction of hull coolers shall be as specified in table I for type A coolers and table II for type B coolers. They shall be used in the construction of the coolers, except that the contractor shall have the option of substituting commercial materials having equal or better physical and chemical properties than the materials specified under Federal or Military specifications (see 6.3 and appendix).

TABLE I. Materials for type A coolers.

Parts	Materials	Specifications
Basic cooler tubes	Copper-nickel alloy (90-10 or 70600)	MIL-C-15726 or MIL-T-16420
Tubes or tube elements	Copper-nickel alloy (90-10 or 70600)	MIL-C-15726 or MIL-T-15005 1/
Housing	Copper-nickel alloy (90-10 or 70600)	MIL-C-15726
Protective anodes	Zinc, type ZPN	MIL-A-18001
Bolts, studs, nuts 2/ and washers	Copper-nickel alloy (90-10 or 70600)	MIL-C-15726
	Phosphor bronze, Comp A or D	QQ-B-750
Gaskets	Synthetic rubber	MIL-G-1149

1/ The following exceptions to MIL-T-15005 apply:

(1) Requirements for outside diameter, wall thickness, and expansion test are not applicable.
(2) The hydrostatic test pressure shall be the same as that required for the completed cooler.
(3) Requirements for the flattening test are not applicable.

2/ Forms of bolts, studs, and nuts shall be in accordance with MIL-S-1222. Copper alloy in accordance with QQ-C-390 is an acceptable alternate material for nozzle nuts.

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TABLE II. Materials for type B coolers.

Parts	Materials	Specifications
Tubes and headers	Aluminum alloy 5052	QQ-A-225 and QQ-A-225/7
	Aluminum alloy 5086	QQ-A-200 and QQ-A-200/5
Bolts, studs, nuts 1/ and washers	Aluminum alloy 5086	QQ-A-200 and QQ-A-200/5
Gaskets	Synthetic rubber	MIL-G-1149

1/ Forms of bolts, studs, and nuts shall be in accordance with MIL-S-1222.

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, materials, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing their intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Threaded parts. Threaded parts, such as bolts, studs, and nuts shall be in accordance with FED-STD-H28, FED-STD-H28/2, and MIL-S-1222. A number 2 fit shall be used as the general working fit. Bolts and studs shall be not less than 1/2 inch in diameter. Threading of nozzles and support studs shall permit nuts to run down to within 1/2 inch of header or support.

3.4 Welding and brazing. Welding and allied processes shall be in accordance with MIL-STD-278. Brazing shall be in accordance with NAVSEA 0900-LP-001-7000, except that the requirements for use of preinserted brazing rings are applicable only to pipe fittings. Brazing will not be permitted on type B coolers.

3.5 Space and weight. In all designs, space and weight shall be held to a minimum consistent with strength and reliability of the coolers.

3.6 Hydrostatic or air pressure. Coolers shall withstand the hydrostatic or air pressure tests (see 4.4.2).

3.7 Ship attitude and motion. Coolers shall perform satisfactorily under the list, pitch, roll, and trim conditions as specified in MIL-STD-1399, section 301.

3.8 Shock. When specified (see 6.2), the coolers shall pass a shock test in accordance with MIL-S-901, grade A (see 4.3.4).

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3.9 Vibration. When specified (see 6.2), the coolers shall not be damaged or caused to malfunction by either the environmental vibrations or by internally excited vibrations (see 4.3.5).

3.10 Design and construction.

3.10.1 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.10.2 Tubes. Coolers shall be constructed of straight tubes welded or brazed into headers, or passages for the cooled fluid shall be constructed of formed plate similarly secured to headers. Tubes of rectangular section shall be used. Tube wall or formed plate thickness shall be a minimum of 0.062 inch of copper-nickel and 0.090 inch for aluminum.

3.10.3 Headers. Headers shall be not less than 1/8 inch thick.

3.10.4 Temperature. Ambient cooling water temperature shall be 85 degrees Fahrenheit (°F) for design purposes.

3.10.5 Surface cooling. The amount of cooling surface provided in all coolers shall be based on the service specified (see 3.10.12), applying a 10 percent fouling factor to the heat transfer coefficient for clean tubes.

3.10.6 Draining. Provision shall be made for draining the cooler. Drain connections shall not project more than 1/4 inch below the bottom of the headers. To achieve this dimension, 1/8 inch pipe size pipe plugs having straight thread with an O-ring, and with slotted or hex head will be acceptable.

3.10.7 Housing. Housings for coolers shall not be furnished.

3.10.8 Velocity. The design velocity through the tubes shall not exceed 7 feet per second.

3.10.9 Protective anodes for type A coolers. The construction of the cooler shall provide for anodes (zincs) as a safeguard against galvanic corrosion. Anodes shall be installed in such a way as to permit ready disassembly from the cooler for cleaning and replacement. The anodes shall be in the form of 1/2 inch thick plate. Each anode shall be secured against bosses on the cooler header, the face of each boss having a minimum of two sharp concentric ridges at least 1/16 inch high machined on them. The zinc shall be secured by a washer and tap bolt. The face of the washer toward the zinc shall be ridged as described above. Two such fastenings may be used per anode where desirable. An anode shall be provided at each end of the cooler.

3.10.10 Gaskets. Gaskets shall be at least 1/8 inch thick. Gaskets shall be provided for the joints between boat or ship hull and tops of headers and center supports, and also inside the hull under the nozzle and support stud washers.

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3.10.11 Nozzles, nuts, and washers. Hexagonal nuts and separate washers shall be furnished for the threaded portion of each nozzle.

3.10.12 Dimensions and performance requirements. Coolers as designated by size shall conform to the following dimensions and performance requirements.

3.10.12.1 Size 100X cooler. The cooler shall be of single pass construction. Nozzles, 4 feet apart center to center, shall lead, at right angles to the plane of the tubes, from the headers. These nozzles shall be threaded with a 1-1/2 inch straight pipe thread to a distance of 4-1/2 inches from the headers. An additional length of the nozzle shall extend 2-1/2 inches from this threaded portion to provide a hose connection. This additional length of nozzle shall be 1-1/2 inches outside diameter (od) and shall be formed out at the end to provide a bead 1-9/16 inches od in order to better secure the hose and clamps. The nozzles shall have a spacer 2 inches od high to provide space between the hull and the cooler. A center support for the cooler shall be provided. To attach this to the boat hull there shall be provided two 3/4 inch diameter studs with nuts and washers. The studs shall be located on a line midway between the nozzles and shall be 4-3/8 inches apart, equidistant from the cooler centerline. The top of the center support shall be in the plane of the tops of the headers. The studs shall extend upward 2 inches from the top of the center support. The overall height of the cooler shall be 1-3/4 inches exclusive of nozzles and studs. The overall length of the cooler shall be 4 feet 3 inches. The overall width of the cooler shall be 8 inches.

(a) The size 100X cooler shall be used for jacket water cooling of an engine with a design point heat rejection of 6,474 British thermal units per minute (Btu/min) and a maximum heat rejection of 10,322 Btu/min.
(b) Fluid to be cooled: fresh water.
(c) Rate of flow of fresh water: 60 gallons per minute (gal/min).
(d) Inlet temperature of fresh water: 185°F.
(e) Outlet temperature of fresh water: 172°F.
(f) Minimum hull speed: 11.8 knots.
(g) Allowable design point pressure drop of fresh water: 7 pounds per square inch (lb/in^2).
(h) Hydrostatic test pressure for cooler: 50 lb/in^2.

3.10.12.2 Size 150 cooler. The cooler shall be of single pass construction. Nozzles, 4 feet 6 inches apart, center to center, shall lead, at right angles to the straight pipe thread to a distance of 2-3/8 inches from the headers. An additional length of nozzle shall extend 3 inches from this threaded portion to provide a hose connection. This additional length of nozzle shall be 2 inches od and shall be formed out at the end to provide a bead 2-1/16 inches od, in order to better secure hose and clamps. A center support for the cooler shall be provided. To attach this to the boat hull there shall be provided two 1/2-inch diameter studs with nuts and washers. These studs shall be located on a line midway between the nozzles and shall be 5-5/8 inches apart, equidistant from the cooler centerline. The top of the center support shall be in the plane of the tops of the headers. The studs shall extend upward 2-1/4 inches from the top of the

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center support. The overall height of the cooler shall be 1-15/16 inches exclusive of nozzles, studs, and zinc anodes. The zinc anodes and fastenings may extend an additional 1-5/16 inches below the bottom of the cooler. The overall length of the cooler shall be 4 feet 9-1/2 inches exclusive of zinc anodes. The overall width of the cooler shall be 7-1/16 inches.

(a) The size 150 cooler shall be used for jacket water cooling of an engine with a design point heat rejection of 10,582 Btu/min and a maximum heat rejection of 12,500 Btu/min.
(b) Fluid to be cooled: fresh water.
(c) Rate of flow of fresh water: 85 gal/min.
(d) Inlet temperature of fresh water: 180°F.
(e) Outlet temperature of fresh water: 165°F.
(f) For design purposes, assume a boat speed of 7-1/2 knots at full power operation.
(g) Allowable design point pressure drop of fresh water: 5 lb/in^2.
(h) Hydrostatic test pressure for cooler: 50 lb/in^2.

3.10.12.3 Size 300X cooler. This cooler shall be of single pass construction. Nozzles, 7 feet apart, center to center, shall lead, at right angles to the plane of the tubes, from the headers. These nozzles shall be threaded with a 2-1/2 inch straight pipe thread for a distance of 4-1/2 inches from the headers. An additional length of nozzle shall extend 2-1/2 inches from this threaded portion to provide a hose connection. This additional length of nozzle shall be 2-1/2 inches od and shall be a bead 2-9/16 inches od at its end, in order to better secure the hose and clamps. The nozzles shall have a spacer 2 inches high to provide space between the hull and the cooler. Two intermediate supports for the cooler shall be provided, located 2 feet from the respective nozzle centers, and 3 feet apart. Each of these intermediate supports shall be fitted with two 3/4-inch diameter studs with nuts and washers for attaching to the boat hull. These two studs shall be 5 inches apart, equidistant from the cooler centerline. The tops of the intermediate supports shall be in the plane of the tops of the headers. The studs shall extend upwards 2 inches from the tops of the supports. The overall height of the cooler shall be 2-3/4 inches exclusive of nozzles and studs. The overall width of the cooler shall be not greater than 9-1/4 inches.

(a) The size 300X cooler shall be used for jacket water cooling of an engine with a design point heat rejection of 16,600 Btu/min and a maximum heat rejection of 22,241 Btu/min.
(b) Fluid to be cooled: fresh water.
(c) Rate of flow of fresh water: 100 gal/min.
(d) Inlet temperature of fresh water: 180°F.
(e) Outlet temperature of fresh water: 160°F.
(f) Minimum hull speed: 11.8 knots.
(g) Hydrostatic test pressure: 35 lb/in^2.
(h) Allowable design point pressure drop of fresh water: 1 lb/in^2.

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3.10.12.4 Size 301 cooler. The cooler shall be of single pass construction. Nozzles, 7 feet apart, center to center, shall lead, at right angles to the plane of the tubes, from the headers. These nozzles shall be threaded with a 2-1/2 inch straight pipe thread for a distance of 2-1/4 inches from this threaded portion to provide a hose connection. This additional length of nozzle shall be 2-1/2 inches od and shall have a bead 2-9/16 inches od at its end, in order to better secure the hose and clamps. Two intermediate supports for the cooler shall be provided, located 2 feet from the respective nozzle centers, and 3 feet apart. For attaching to the boat hull, each of these intermediate supports shall be fitted with two 5/8-inch diameter studs with nuts and washers. Two studs on each support shall be 5-5/8 inches apart, equidistant from the cooler centerline. The tops of the intermediate supports shall be in the plane of the tops of the headers. The studs shall extend upwards 2-1/4 inches from the tops of the supports. The overall height of the cooler shall be 2-3/4 inches exclusive of nozzles, studs, and zinc anodes. The zinc anodes with their bosses and fastenings may extend an additional 1-5/16 inches below the bottom of the cooler. The overall length of the cooler shall be 7 feet 4 inches exclusive of the zinc anodes. The anodes may extend an additional 3/8 inch at either end if desirable. The overall width of the cooler shall be not greater than 7-1/6 inches.

(a) The size 301 cooler shall be used for jacket water cooling of an engine with a design point heat rejection of 16,000 Btu/min and a maximum heat rejection of 22,241 Btu/min.
(b) Fluid to be cooled: fresh water.
(c) Rate of flow of fresh water: 100 gal/min.
(d) Inlet temperature of fresh water: 180°F.
(e) Outlet temperature of fresh water: 160°F.
(f) For design purposes assume a boat speed of 6-1/2 knots at full power operation.
(g) Allowable design point pressure drop of fresh water: Less than 5 lb/in^2. Pressure drop shall not exceed 5 lb/in^2 when flow rate is increased to 162 gal/min.
(h) Hydrostatic test pressure for cooler: 35 lb/in^2.

3.10.12.5 Size 400 cooler. The cooler shall be of three pass construction. Nozzles, 7 feet apart, center to center, shall lead at right angles to the plane of the tubes, from a pan 6-11/32 inches from one side of the cooler (nozzles are not on the centerline of the cooler) and shall have a height of 5-1/2 inches. These nozzles shall be threaded with a 2-1/2 inch pipe thread for a distance of 3 inches. An additional length of the nozzle shall extend 2-1/2 inches from the thread portion to provide a hose connection. This additional length of the nozzle shall be 2-1/2 inches od and shall have a bead of 2-9/16 inches od at its end in order to better secure the hose and clamps. Two intermediate supports for the cooler shall be provided, located 2 feet from the respective nozzle center, and 3 feet apart. For attaching to the boat hull, each of these intermediate supports shall be fitted with 5/8-inch diameter studs with nuts and washers. The two studs on each support shall be 5-5/8 inches apart, equidistant from the centerline of the nozzles. The top of the intermediate supports shall be in the plane of the top of the headers. The studs shall extend upward 5-1/2 inches from the tops of the supports. The overall height of the coolers shall be 2-3/4 inches exclusive of nozzles, pans, studs, and zinc anodes. The zinc anodes with their bosses and fastenings may extend an additional 1-5/16 inches below the bottom of the cooler. The overall width of the cooler shall be not greater than 13-5/8 inches.

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(a) The size 400 cooler shall be used for jacket water cooling of an engine with a design point heat rejection of 18,583 Btu/min and a maximum heat rejection of 26,632 Btu/min.
(b) Fluid to be cooled: fresh water.
(c) Rate of flow of fresh water: 60 gal/min.
(d) Inlet temperature of fresh water: 172°F.
(e) Outlet temperature of fresh water: 134.7°F.
(f) Minimum hull speed: 6.5 knots.
(g) Hydrostatic test pressure: 35 lb/in^2.

3.10.12.6 Size 500 cooler. This cooler shall be of two-pass construction. The inlet and outlet nozzles shall be located 3-5/8 inches from one end of the cooler, shall be 7-1/2 inches apart equidistant (3-3/4 inches) from the cooler centerline, and shall lead from the header at right angles to the plane of the tubes. These nozzles shall be threaded with a 2-1/2 inch straight pipe thread to a distance of 3-1/4 inches from the header. An additional length of each nozzle shall be extended 1-1/2 inches from the threaded portion to provide a hose connection. This additional length of nozzle shall be 2-1/2 inches od and shall have a bead 2-9/16 inches od in order to better secure the hose and clamps. On the opposite end of the cooler there shall be two studs 1 inch in diameter, 3-5/8 inches from the end. The studs shall have nuts and washers. These studs shall be 7-1/2 inches apart equidistant (3-3/4 inches) from the cooler centerline. The studs shall extend upward 3-1/4 inches from the top of the header. The overall height of the cooler shall be 2-3/4 inches exclusive of nozzles and studs. The overall length of the cooler shall be 79-1/4 inches. The overall width of the cooler shall be 14-9/16 inches. An intermediate support plate with two studs shall be provided. The studs shall be 3/4 inch in diameter and 3-1/4 inches long, and shall be 9-3/8 inches apart equidistant (4-11/16 inches) from the cooler centerline. The studs shall have nuts and washers. There shall be a 3/4-inch thick neoprene gasket on both headers and the intermediate support plate. There shall be a 1/4-inch neoprene gasket for all studs and nozzles. Neoprene shall be 60 durometer. There shall be four drain plugs, two on each header.

(a) The size 500 cooler shall be used for a jacket water cooling of an engine with a design point heat rejection of 4,385 Btu/min and a maximum heat rejection of 5,900 Btu/min for the after cooler circuit; heat rejection of design point 25,470 Btu/min and maximum heat rejection of 39,950 Btu/min for the engine cooling circuit.
(b) Fluid to be cooled: fresh water.
(c) Rate of flow of fresh water:
100 gal/min for the aftercooler circuit.
142 gal/min for the engine cooling circuit.
(d) Inlet temperature of fresh water:
100.3°F for aftercooler circuit.
170°F for the engine cooling circuit.
(e) Outlet temperature of fresh water:
95°F for the aftercooler circuit.
148°F for the engine cooling circuit.

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(f) Minimum hull speed: 10 knots.
(g) Hydrostatic test pressure: 35 lb/in^2.
(h) Allowable design point pressure drop of fresh water:
1.4 lb/in^2 for the aftercooler.
3.2 lb/in^2 for the engine cooling circuit.

3.10.12.7 Size 1000 cooler. This cooler shall be either single-pass or two-pass construction. It shall be either type A or type B. It may be of a special configuration to suit the hull of the ship.

3.11 Manufacturer's service part number. Where a complete part numbering system is not in use, the manufacturer shall assign service part numbers to each cooler and to each other part or subassembly which in his judgment may require replacement during the life of the unit.

3.12 Identification of coolers. Coolers shall be permanently marked with data which shall include manufacturer's name, service part number, date of manufacture, and National Stock Number (NSN). This information may be stamped on the cooler in lieu of providing an identification plate.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. A sample cooler of each size covered in the contract or order shall be inspected as specified in 4.3.1 through 4.3.5 (see 6.3).

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4.3.1 Performance test. A performance test shall be conducted on the cooler under the design conditions of flow, removal, and temperatures of cooling medium.

4.3.2 Pressure drop test. The cooler shall be tested to determine pressure drop under various flow conditions (see 3.10.12).

4.3.3 Determined weight. Construction, materials, and workmanship shall be examined and weight shall be determined.

4.3.4 Shock test. When specified (see 6.2), a shock test shall be performed in accordance with MIL-S-901, grade A (see 6.3).

4.3.5 Vibration test. When specified (see 6.2), a vibration test shall be performed in accordance with MIL-STD-167-1, type I and type II (see 6.3).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests of 4.4.1 through 4.4.2.2 (see 6.3).

4.4.1 Examination. Each cooler offered for delivery shall be examined for alignment, fit, material, finish, and conformance to the requirements of this specification not requiring tests.

4.4.2 Hydrostatic or air pressure test. Each cooler offered for delivery shall be hydrostatically or air pressure tested as specified in 4.4.2.1 or 4.4.2.2.

4.4.2.1 Hydrostatic test. Each cooler shall be given a hydrostatic test. The test pressure shall be as specified for the particular size of cooler (see 3.10.12).

4.4.2.2 Air pressure test. An air pressure test in which the cooler is subjected to an internal pressure equal to that specified for hydrostatic testing, and is submerged in water, may be substituted for the hydrostatic test.

4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The coolers shall be preserved and packaged in accordance with method III of MIL-P-116. Openings shall be sealed with noncorrosive materials in a manner that shall prohibit the entrance of foreign matter.

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5.1.2 Commercial. Preservation shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.2).

5.2.1 Level A. Each cooler preserved as specified (see 5.1) shall be packed for overseas shipment in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Overseas type
PPP-B-621	Class 2
MIL-C-104	Optional

When required, shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Shipping containers shall be closed, strapped, or banded in accordance with the applicable box specification or appendix thereto. When the gross weight of wood or wood-cleated boxes exceeds 200 pounds, wood skids in accordance with the applicable box specification shall be applied.

5.2.2 Level B. Each cooler preserved as specified (see 5.1) shall be packed for domestic shipment and storage in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Domestic
PPP-B-621	Class 1
MIL-C-104	Optional

Shipping containers shall be closed in accordance with the applicable box specification or appendix thereto. When the gross weight of wood and wood-cleated boxes exceeds 200 pounds, wood skids in accordance with the applicable box specification shall be applied.

5.2.3 Commercial. Each cooler preserved as specified shall be packed in accordance with ASTM D 3951.

5.3 Cushioning, blocking, and bracing. Cushioning, blocking, and bracing shall be in accordance with MIL-STD-1186, except that excelsior and shredded newspaper shall not be used.

5.4 Marking. In addition to any special marking required (see 6.2), level A interior pack and level A and B exterior shipping containers shall be marked in accordance with MIL-STD-129; commercial packs and shipping containers shall be marked in accordance with ASTM D 3951.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coolers covered by this specification are intended for use as engine jacket water coolers for Naval ships and boats.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Type and size cooler required (see 1.2).
(d) If coolers are to be other than type A (see 1.2).
(e) If first article is required (see 3.1).
(f) Identification of coolant and fluid to be cooled.
(g) Rate of flow of fluid to be cooled.
(h) Inlet temperature of fluid to be cooled.
(i) Outlet temperature of fluid to be cooled or rate of heat exchange.
(j) Velocity and number of passes of cooled fluid through cooler.
(k) Working pressure of cooled fluid.
(l) Type, manufacturer, and model number of engine to be cooled.
(m) Rating of engine in horsepower or kilowatts at revolutions per minute.
(n) Application of engine (propulsion, bowthruster, generator, and so forth).
(o) Type of ship and service, and hull construction (steel, fiberglass, wood or aluminum).
(p) For propulsion engines, minimum hull speed at full power.
(q) For generator set engines, sized for 0-knot hull speed depending on a thermal or convection current of 1/2 knot, specify if other.
(r) Maximum sea water temperature.
(s) Percent of glycol in jacket water circuit.
(t) When shock test is required (see 3.8 and 4.3.4).
(u) When vibration test is required (see 3.9 and 4.3.5).
(v) Level of preservation and packing required (see 5.1 and 5.2).
(w) Special marking required (see 5.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2 and appendix	DI-DRPR-80651	Engineering drawings	-----
4.3	DI-T-4902	First article inspection report	-----
4.3.4	UDI-T-23753	Reports, equipment shock test	-----
4.3.5	UDI-T-23762	Report, vibration testing	-----
4.4	DI-T-5329	Inspection and test reports	-----

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first _____ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions. As used herein, "contracting agency" refers to the activity purchasing the equipment covered by this specification. The contracting activity may be the Government or a shipbuilder. The "design review agency" refers to the activity responsible for controlling the technical requirements for design, generally a Government command or agency such as NAVSEA or an authorized representative. Communication with the design review agency should be handled through the contracting activity.

6.6 Sub-contracted material and parts. The packaging requirements of reference documents listed in section 2 do not apply when materials and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.7 Subject term word key listing.

Cooling
Diesel engines
Heat exchangers
Heat transfer
Keel coolers

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
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APPENDIX

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings. When required by the contract or order, drawings shall contain the following information:

30.1.1 Installation control drawing. An installation control drawing (familiarly known as an outline drawing) shall show necessary external views of the cooler assembly, and shall include the external dimensions necessary for reproduction on a ship's hull drawing. The main purpose of the installation control drawing is to show the interfaces, including piping, between the hull and the cooler. The drawing shall show the complete dimensions for installation, the size of all openings, and location of zinc anodes.

30.1.2 Assembly drawing. An assembly drawing showing longitudinal and transverse cross-section views of the cooler assembly shall also show the relationship of parts, the arrangement of tubes in the cooler, support plates, and weldments of adjoining parts. Adequate use shall be made of enlarged views or sections. If necessary, subassembly drawings conforming to the above may be furnished for components of the cooler. The assembly drawing or drawings shall be such that a thorough understanding of the design and construction may be obtained without reference to related detail drawings.

30.1.2.1 List of materials. The assembly drawing shall contain a complete list of materials to show the names and corresponding identifying numbers and materials of the parts used in construction of the cooler. The identifying part number shall also be shown adjacent to the part depicted in the various views, with arrows pointing to the parts. If the manufacturer exercises the option as specified in 3.2, he shall include two columns in the "List of Materials" on his drawings; one column shall be used for the specified Federal or Military specifications, and the other shall be used for the commercial specification number for the substituted material. Notes shall then appear on the drawings stating that:

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(a) The design is based on the Federal or Military specification materials.
(b) The materials listed in the commercial specifications column have physical and chemical properties which are equal to or better than the specified Federal or Military specifications and may be substituted by the manufacturer.

30.1.2.1.1 Corrosion prevention anodes. The assembly drawing shall show the size and location of the corrosion prevention anodes.

30.1.2.1.2 Detail assembly drawings. The assembly drawing shall include detail assembly drawings of major parts such as tubes, headers, and support plates, all of which shall be accurately dimensioned. Finishes, welding symbols, and identifying part numbers shall be indicated as necessary for manufacture.

30.1.3 Special purpose drawing. One drawing shall be a special purpose drawing titled: "Drawing List and Performance Data for Hull _____". (Type of cooler, for example, Hull Cooler, and its designation by model number or by service part number to be entered to complete the title.)

30.1.3.1 The "Drawing list" tabulation shall include the following columns:

(a) Drawing title.
(b) Manufacturer's drawing number.
(c) Revision symbol.

The list shall include all master drawings which comprise the given design. The revision symbol column shall be kept up-to-date to the time of manufacture so that it will indicate the latest revision of each drawing applicable to the equipment as built.

30.1.3.1.1 The "Performance data" tabulation shall include the following information:

(a) Identification of cooled medium.
(b) Flow rate of cooled medium (gal/min).
(c) Inlet temperature of cooled medium (°F).
(d) Outlet temperature of cooled medium (°F).
(e) Pressure drop of cooled medium through cooler (lb/in^2).
(f) Velocity of cooled medium at inlet connection (feet per second (ft/s)).
(g) Velocity of cooled medium through cooler (ft/s).
(h) Number of passes of cooled medium through cooler.
(i) Working pressure of cooled medium (lb/in^2 gauge).
(j) Test pressure (lb/in^2 gauge).
(k) Logarithmic mean temperature difference (°F).
(l) Heat transfer (British thermal units (Btu) per hour per square foot (ft^2) per °F logarithmic mean temperature difference).

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(m) Cooling surface (ft^2).
(n) Heat transfer capacity at design point (Btu per hour).
(o) Ship speed in knots at design point (omit if figure is classified).
(p) Ship minimum speed at full power (knots).

30.1.3.1.2 The certification data shall include notes indicating the contract or order number, the application (service) of the unit, the number of coolers per ship and per unit of parent equipment (that is, per engine), and dry and wet weights of the cooler. The following examples illustrate the type of notes required:

"(a) ABC Engine Company Purchase order 5678 on EFG Cooler Mfg. Co.
"(b) Service: Jacket water cooling for main propulsion diesel engine.
"(c) Two coolers per ship, one cooler per engine, or 12 coolers furnished as stock material.
"(d) Dry weight of cooler _____ pounds; wet weight of cooler _____ pounds."

30.1.3.1.3 Above the title block of the certification data sheet there shall be entered the designations of the applicable vessels or ships.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-16388D(SH)	COOLERS, FLUID, NAVAL SHIPBOARD (HULL COOLERS)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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